Lecture 4

Lecture 4 drops the microscopic input. We state what is required of a CFT_2 for a semiclassical AdS_3 dual — large c, few light operators, factorised correlators, a gap in the higher-spin spectrum, and a sparse light spectrum — and prove the theorem of Hartman, Keller and Stoica that the last of these, with modular invariance, already forces the free energy and the density of states of $3d$ gravity at all temperatures and for all $\Delta \geq c/6$. We close by computing the gravity side from the action: the Gibbons–Hawking term, the counterterm, the boundary stress tensor in Fefferman–Graham gauge, the masses and angular momenta, and the renormalised on-shell actions whose competition is the Hawking–Page transition.

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Conventions

Signature $(-,+,+)$, $\Lambda = -1/\ell^2$, $\hbar = c_{\text{light}} = 1$. The BTZ black hole is

$$\mathrm{d}s^2 = -f\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f} + r^2\big(\mathrm{d}\varphi + N^\varphi \mathrm{d}t\big)^2, \qquad f = \frac{(r^2 - r_+^2)(r^2 - r_-^2)}{\ell^2 r^2}, \qquad N^\varphi = -\frac{r_+ r_-}{\ell r^2}, \tag{1}$$

with $\varphi \simeq \varphi + 2\pi$ and

$$M = \frac{r_+^2 + r_-^2}{8G\ell^2}, \quad J = \frac{r_+ r_-}{4G\ell}, \quad S = \frac{2\pi r_+}{4G}, \quad T = \frac{r_+^2 - r_-^2}{2\pi \ell^2 r_+}, \quad \Omega_H = \frac{r_-}{\ell r_+}, \quad c = \bar{c} = \frac{3\ell}{2G}. \tag{2}$$

The dual CFT_2 lives on the cylinder $\{(t,\varphi)\}$ at spatial infinity, of spatial circumference $2\pi\ell$. We measure time in units of ℓ, i.e. we use

$$x^{\pm} = \frac{t}{\ell} \pm \varphi, \tag{3}$$

so that the boundary circle has circumference 2π and all CFT quantities are dimensionless. Two sets of variables appear throughout and we fix them here.

Conformal weights. $\Delta_L \equiv L_0$ and $\Delta_R \equiv \bar{L}_0$ are the weights measured from the CFT vacuum, $\Delta = \Delta_L + \Delta_R$, spin $s = \Delta_L - \Delta_R$. The vacuum has $\Delta = 0$.

Energies. $E_L \equiv L_0 - \frac{c}{24}$, $E_R \equiv \bar{L}_0 - \frac{c}{24}$, $E = E_L + E_R = \Delta - \frac{c}{12}$. These are the energies conjugate to the modular-invariant inverse temperatures below, and the variables in which the gravitational formulas are simple. The dictionary to the bulk is

$$E_L = \frac{\ell M + J}{2} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad E_R = \frac{\ell M - J}{2} = \frac{(r_+ - r_-)^2}{16G\ell}, \tag{4}$$

[checked symbolically] so that $\ell M = E_L + E_R$ and $J = E_L - E_R$. For the non-rotating representatives:

	$E_L = E_R$	Δ	bulk
global AdS_3	$-c/24$	0	CFT vacuum, $M = -1/8G$
conical defects	$-\frac{c}{24n^2}$	$\frac{c}{12}\left(1 - \frac{1}{n^2}\right)$	$\mathbb{Z}_n$ defect, $M = -1/8Gn^2$
$M = J = 0$ geometry	0	$c/12$	Ramond ground state
Hawking–Page	$c/24$	$c/6$	$r_+ = \ell$

[checked]. The last line is derived in §2.4; the first three are from lecture 2, §2.3 and [1, 2].

Inverse temperatures. The (dimensionless) potentials conjugate to E_L, E_R are

$$\beta_L = \frac{2\pi\ell}{r_+ + r_-}, \qquad \beta_R = \frac{2\pi\ell}{r_+ - r_-}, \qquad \hat{\beta} \equiv \frac{\beta_{\mathrm{phys}}}{\ell} = \frac{\beta_L + \beta_R}{2}, \qquad \hat{\beta}\,\hat{\Omega} = \frac{\beta_R - \beta_L}{2}, \tag{5}$$

with $\hat{\Omega} = \ell\Omega_H$ [checked symbolically]. In terms of the Euclidean modular parameter of the boundary torus, $\beta_L = -2\pi i\tau$, $\beta_R = 2\pi i\bar{\tau}$, so that

$$Z(\beta_L, \beta_R) = \mathrm{Tr}\, e^{-\beta_L E_L - \beta_R E_R} = \mathrm{Tr}\, q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{c}{24}}, \qquad q = e^{2\pi i\tau} = e^{-\beta_L}, \tag{6}$$

(For $\beta_L \neq \beta_R$ the two are independent: τ and $\bar{\tau}$ are both purely imaginary but $\bar{\tau} = \tau^*$ only when $\beta_L = \beta_R$. See [10, fn. 3].) The modular S transformation $\tau \to -1/\tau$ acts as

$$\beta_L \;\to\; \beta_L' = \frac{4\pi^2}{\beta_L}, \qquad \beta_R \;\to\; \beta_R' = \frac{4\pi^2}{\beta_R}. \tag{7}$$

The self-dual point is $\beta_L = \beta_R = 2\pi$, i.e. $\beta_{\mathrm{phys}} = 2\pi\ell$, the Hawking–Page temperature of lecture 2, §1.6.

String theory. Type IIB with $16\pi G_{10} = (2\pi)^7 g_s^2 \alpha'^4$, i.e. $G_{10} = 8\pi^6 g_s^2 \alpha'^4$. The volume of a compact T^4 is written $V = (2\pi)^4 v\, \alpha'^2$ with v dimensionless, so that the six-dimensional string coupling is $g_6^2 = g_s^2/v$.

1 What makes a CFT_2 holographic

Every example in lecture 3 was obtained by starting from a brane configuration and computing. We now reverse the question: given a CFT_2, what properties must it have for a semiclassical AdS_3 dual to exist? The statements we actually want to make about $3d$ gravity — the entropy of a general BTZ black hole, the free energy, the two-point functions, entanglement entropy — do not depend on which string compactification one starts from, so it should be possible to derive them without one.

1.1 The El-Showk–Papadodimas conditions

El-Showk and Papadodimas [7] propose that a CFT is naturally described holographically in AdS if it has three properties:

(i) large central charge;

(ii) a small number of operators of low conformal dimension;

(iii) factorisation of the correlators of those low-lying operators.

We take them in turn, and check the logical relation between them.

(i) Large c

In AdS_3 this is Brown–Henneaux read backwards:

$$c = \frac{3\ell}{2G} \qquad \Longleftrightarrow \qquad \frac{G}{\ell} = \frac{3}{2c}, \tag{8}$$

so the semiclassical limit $G/\ell \to 0$ *is* $c \to \infty$. There is no other parameter: with $\hbar = 1$ Newton's constant in three dimensions has dimensions of length, so $\ell/G \sim c$ is the only dimensionless combination available. The bulk loop-counting parameter is $G/\ell \sim 1/c$, and the coupling constant at the level of amplitudes, as El-Showk and Papadodimas note [7, §7.1], is

$$g = \frac{1}{\sqrt{c}}. \tag{9}$$

(ii) Few light operators

In a CFT_2 this must be phrased with care, because Virasoro descendants are automatically light: the vacuum module contains $L_{-2}|0\rangle$ at $\Delta = 2$, $L_{-3}|0\rangle$ at $\Delta = 3$, $L_{-2}^2|0\rangle$ and $L_{-4}|0\rangle$ at $\Delta = 4$, and so on, with degeneracy at level n equal to the number of partitions of n into parts ≥ 2. Those states are the boundary gravitons of lecture 2, §5 and they must be present. The condition is instead about *Virasoro primaries*:

$$\#\{\text{Virasoro primaries with } \Delta = \mathcal{O}(c^0)\} = \mathcal{O}(c^0). \tag{10}$$

Pure $3d$ gravity is the extreme case: no primaries besides the vacuum, so the entire light spectrum is the vacuum Virasoro module. A supergravity dual with n light bulk fields has n light primaries. The free symmetric orbifold has infinitely many (§2.3 (lecture 3)), but still only $\mathcal{O}(c^0)$ at each fixed Δ — which is why it satisfies (ii) yet fails the stronger conditions of §1.2 and §1.3.

(iii) Factorisation, and why it forces large c

Normalise the light primaries so that $\langle \mathcal{O}_i(x)\mathcal{O}_j(0)\rangle = \delta_{ij}|x|^{-2\Delta_i}$. Factorisation is the statement that connected correlators are suppressed:

$$\boxed{\big\langle \mathcal{O}_1\mathcal{O}_2\cdots\mathcal{O}_n\big\rangle_{\text{connected}} \;=\; \mathcal{O}\Big(c^{-\frac{n-2}{2}}\Big)\,,} \tag{11}$$

so that at leading order in $1/c$ the $\mathcal{O}_i$ are *generalised free fields*: their correlators reduce to sums of products of two-point functions, the spectrum of their composites is a Fock space, and they are naturally represented as free fields in one dimension more. In particular

$$C_{ijk} = \langle \mathcal{O}_i\mathcal{O}_j\mathcal{O}_k\rangle \;\sim\; \frac{1}{\sqrt{c}}, \tag{12}$$

which in the bulk says that a cubic vertex carries a factor of $\sqrt{G}$.

Condition (iii) implies condition (i) [7, §4.4], by the following argument. Consider $\langle\mathcal{O}\mathcal{O}\mathcal{O}\mathcal{O}\rangle$ for a single light primary of dimension $\Delta_{\mathcal{O}}$. Whatever else appears in the $\mathcal{O}\times\mathcal{O}$ operator product expansion, the stress tensor certainly does, with a coefficient fixed by the conformal Ward identity:

$$\mathcal{O}(z)\mathcal{O}(0) \;\sim\; \frac{1}{z^{2\Delta_{\mathcal{O}}}}\Big[1+\frac{2\Delta_{\mathcal{O}}}{c}\,z^2\,T(0)+\dots\Big]. \tag{13}$$

Its contribution to the four-point function is the stress-tensor conformal block with coefficient

$$\frac{C^2_{\mathcal{O}\mathcal{O}T}}{\langle TT\rangle} = \frac{2\Delta_{\mathcal{O}}^2}{c}, \tag{14}$$

[checked], and this sits in the *connected* part. Factorisation (11) at $n=4$ therefore requires $\Delta_{\mathcal{O}}^2/c \ll 1$. Since the stress tensor is present in every CFT and cannot be switched off, factorisation for light operators is impossible at finite c: one needs a family of theories with $c\to\infty$. This is the sense in which the bulk is an approximation rather than a rewriting. It also means the light sector cannot be the whole theory: a theory containing only the generalised-free-field states would have an $\mathcal{O}(c^0)$ free energy, whereas $c\to\infty$ requires $\mathcal{O}(c)$ degrees of freedom. The missing states are the black hole microstates.

1.2 The gap

Conditions (i)–(iii) give *some* local bulk theory. They do not give *Einstein* gravity. The extra requirement, due to Heemskerk, Penedones, Polchinski and Sully [8] and sharpened for CFT_2 in [9], is a gap in the spectrum of single-trace higher-spin operators ("single-trace" means single-particle in the bulk; in a symmetric orbifold it means single-cycle):

$$\Delta_{\text{gap}} \equiv \min\{\Delta:\ \mathcal{O}\text{ single-trace primary with spin } s>2\} \;\longrightarrow\; \infty. \tag{15}$$

In the bulk, a single-trace operator of spin s is a string state on the leading Regge trajectory, with $m^2\sim s/\alpha'$, and for large mass $\Delta\simeq m\ell$; hence

$$\Delta_{\text{gap}} \;\sim\; \frac{\ell}{\ell_s}, \tag{16}$$

and (15) says the string scale is far below the AdS scale. Massive string states then decouple, the low-energy bulk action is Einstein gravity plus a finite number of light fields, and corrections

are organised in $1/\Delta_{\rm gap}$. Without (15) the bulk graviton three-point function is not the Einstein one: it contains the two additional structures allowed by conformal invariance, with coefficients suppressed by $\Delta_{\rm gap}^{-2}$ and $\Delta_{\rm gap}^{-4}$ [9].

Every example of lecture 3 fails (15) at the orbifold point: $\Delta_{\rm gap} = \mathcal{O}(1)$, because the currents (51) (lecture 3) have $\Delta = s$ exactly. Restoring the gap is what the twist-two deformation (53) (lecture 3) is supposed to do.

1.3 Sparseness of the light spectrum

Conditions (i)–(iii) and (15) constrain *operators*. For thermodynamics — black holes — what matters is the *density* of light states. The relevant condition was isolated by Hartman, Keller and Stoica [10], and was already forced on us by (38) (lecture 3):

A family of CFT_2*'s with* $c \to \infty$ *has a* ***sparse light spectrum*** *if*

$$\boxed{\rho(\Delta) \;\lesssim\; e^{2\pi\Delta} \qquad \text{for} \qquad \Delta \le \frac{c}{12}.} \tag{17}$$

With an angular potential one needs in addition

$$\boxed{\rho(\Delta_L,\Delta_R) \;\lesssim\; e^{4\pi\sqrt{\Delta_L\Delta_R}} \qquad \text{for} \qquad \Delta_L < \frac{c}{24} \;\text{ or }\; \Delta_R < \frac{c}{24}.} \tag{18}$$

These are [10, eq. (1.10) and (1.15)], written in terms of $\Delta_{L,R} = E_{L,R} + c/24$. Setting $\Delta_L = \Delta_R = \Delta/2$ in (18) returns $e^{2\pi\Delta}$, so the two are consistent.

Why the exponent is 2π

The number 2π is not adjustable: it is the self-dual inverse temperature of (7). The vacuum contributes $e^{\pi c/6}$ to $Z(\beta = 2\pi)$, and a light spectrum growing faster than $e^{2\pi\Delta}$ would give

$$\sum_{\Delta \le c/12} \rho(\Delta)\, e^{-2\pi\Delta} \;\gg\; 1, \tag{19}$$

swamping it and destroying the vacuum dominance on which (33) (lecture 3), and hence Cardy's formula, rests. Section 2.1 shows that (17) is not only sufficient but necessary for the free energy to be the gravitational one.

Hagedorn and sub-Hagedorn

A density of states growing as $\rho \sim e^{\beta_H \Delta}$ is by definition Hagedorn behaviour, with β_H the inverse Hagedorn temperature: the canonical sum $\sum \rho(\Delta) e^{-\beta\Delta}$ diverges for $\beta < \beta_H$. So (17) says that light states may grow at most Hagedorn-fast, with $\beta_H = 2\pi$, which is precisely the Hawking–Page temperature. A theory saturating (17) has a string at the AdS scale. A theory with a semiclassical supergravity dual should grow strictly more slowly, and it is useful to know what "more slowly" means quantitatively.

A gas in the bulk. Suppose the bulk contains a finite number of free fields. The light spectrum of the boundary theory is then the multi-particle Fock space built on the single-particle states, i.e. a gas. Let the single-particle density of states grow as a power, $d_1(\Delta) \simeq a\,\Delta^p$. The multi-particle (bosonic) partition function is

$$\log Z = -\sum_{\Delta} d_1(\Delta)\log\!\left(1-e^{-\beta\Delta}\right) = \sum_{m\geq 1}\frac{1}{m}\int \mathrm{d}\Delta\; a\Delta^p e^{-m\beta\Delta} = \frac{a\,\Gamma(p+1)\,\zeta(p+2)}{\beta^{\,p+1}}, \tag{20}$$

[checked], hence $E = -\partial_\beta \log Z \propto \beta^{-(p+2)}$ and $S = (1-\beta\partial_\beta)\log Z \propto \beta^{-(p+1)}$, so

$$\boxed{\log\rho(E) \;\sim\; E^{\frac{p+1}{p+2}} \;=\; E^{\frac{D-1}{D}}, \qquad D \equiv p+2,} \tag{21}$$

the familiar statement that a gas of massless particles in D spacetime dimensions has $S \sim E^{(D-1)/D}$ [checked]. Two cases:

- *A single bulk field in* AdS_3. Its boundary descendants are $\partial^{n_1}\bar\partial^{n_2}\mathcal{O}$, so $d_1(\Delta) \simeq \Delta$: $p=1$, $D=3$, and $\log\rho \sim \Delta^{2/3}$.
- *Boundary gravitons only (pure gravity).* The single-particle states are the generators L_{-n}, one per level, so $d_1 = \mathcal{O}(1)$: $p=0$, $D=2$, and $\log\rho \sim \Delta^{1/2}$ — equivalently, Cardy with $c_{\text{eff}} = 1$ per chirality.

For any finite p, (21) is strictly sub-Hagedorn: $\Delta^{(D-1)/D} \ll 2\pi\Delta$ at large Δ. Hagedorn growth requires the single-particle density itself to grow exponentially, i.e. a tower of string states. This is the precise sense in which "sparse" means "no string at the AdS scale".

The landscape. Belin, Castro, Gomes, Keller and Mühlmann [16, 17, 18] made this dichotomy quantitative for symmetric-product orbifolds by studying the growth of the Fourier coefficients of their elliptic genera — which, being protected, can be computed at the orbifold point and used everywhere on $\mathcal{M}$. The space of such theories splits into a *fast-growth* region, where the light states grow Hagedorn-fast and the bulk must be stringy, and a *slow-growth* region compatible with a supergravity dual; they construct elliptic genera in the second class, i.e. candidate $\mathrm{AdS}_3/\mathrm{CFT}_2$ pairs whose bulk is semiclassical supergravity rather than a tensionless string.

Remark 1. The importance of controlling the light spectrum was recognised earlier in the modular-bootstrap literature — Hellerman's bound $\Delta_1 \leq c/6 + 0.474\ldots$ [11] (in our conventions, where $c = \bar c$ and $\Delta = \Delta_L + \Delta_R$; see [10, §2.5]) and its refinements [12] — and in the analysis of permutation orbifolds by Belin, Keller and Maloney [13, 14] and by Haehl and Rangamani [15]. What [10] adds is the sharp *if and only if* of §2.1.

1.4 Good and bad candidates

Two examples show that (17) is a strong constraint, failed by most large-c theories.

Bad: the tensor product $\mathcal{C}^{\otimes N}$. Take N decoupled copies of a fixed seed $\mathcal{C}$ of central charge c_1, so $c = Nc_1$, and let $\mathcal{C}$ have $d \geq 1$ primaries at dimension $\Delta_1 = \mathcal{O}(1)$. Exciting m of the copies gives states of dimension $m\Delta_1$ with degeneracy at least $\binom{N}{m}d^m$, and $\binom{N}{m} \geq (N/m)^m$, so

$$\log\rho(m\Delta_1) \;\geq\; m\log\frac{N}{m} + m\log d. \tag{22}$$

Sparseness (17) demands $\log\rho \lesssim 2\pi m\Delta_1$, i.e. $\log(N/m) \lesssim 2\pi\Delta_1 - \log d$, which fails at fixed m once N is large enough [checked]. The tensor product is parametrically too dense at low dimension, and the reason is transparent: it has N independent sectors and therefore N independent ways of being slightly excited. There is no confinement, no Hawking–Page transition, and no gravitational dual. Large c by itself buys nothing.

Marginal: the symmetric orbifold $\mathrm{Sym}^N(\mathcal{C})$**.** The S_N projection removes the $\binom{N}{m}$: an untwisted state with m copies excited must be symmetrised, so the count at fixed m becomes a number of partitions, independent of N. The twisted sectors put states back, and the balance is exact. For *any* seed, HKS show [10, §5] that the density of states saturates (18),

$$\rho(\Delta_L,\Delta_R) \;\sim\; e^{4\pi\sqrt{\Delta_L\Delta_R}} \qquad \text{for} \qquad E_LE_R < \left(\frac{c}{24}\right)^2, \tag{23}$$

and equals the Cardy value above that hyperbola. Symmetric orbifolds are therefore the densest theories compatible with $3d$ gravity: they have the correct leading free energy and the correct Hawking–Page transition, together with a Hagedorn tower of light states. This is the thermodynamic version of §2.3 (lecture 3).

Good, conjecturally: deformed symmetric orbifolds. Switching on the twist-two deformation (53) (lecture 3) gives anomalous dimensions to states that were degenerate at the orbifold point. In a supersymmetric theory states are lifted in pairs: an unprotected state can leave the light spectrum only by joining a long multiplet with a partner. Explicit second-order conformal perturbation theory confirms lifting at low levels [3, 4, 6, 5]. Whether the deformed theory has strictly sub-Hagedorn light growth — and hence, by §1.3, a semiclassical supergravity dual — is expected but, as far as we know, not proved [unverified]; [17] is the systematic attack on the question using protected data.

Good, if it exists: pure gravity. A CFT with no light primaries at all would have $\rho(\Delta)$ given by the vacuum Virasoro character alone, $\log\rho \sim \Delta^{1/2}$: as sparse as possible. It would saturate the *lower* end of the allowed band (46) below, where symmetric orbifolds saturate the upper end. Whether such a CFT exists is the Maloney–Witten problem; see §3.4.

Exercise 1. Take the seed $\mathcal{C}$ to be a single free boson at generic radius, $c_1 = 1$, with $\rho_1(\Delta) \sim e^{2\pi\sqrt{\Delta/3}}$. (i) Verify that this is sub-Hagedorn. (ii) Compute the light-state growth of $\mathcal{C}^{\otimes N}$ and of $\mathrm{Sym}^N(\mathcal{C})$, and locate each relative to (17). (iii) Consider instead the orbifold by a cyclic $\mathbb{Z}_N \subset S_N$. How many twisted sectors are there, and what is the ground-state energy in each? Compare with [13], where permutation orbifolds are shown generically to have a Hagedorn or faster light spectrum.

2 Cardy's formula in the holographic regime

We now show that sparseness is exactly what is needed. The result is due to Hartman, Keller and Stoica [10]; the derivations below follow their paper, filling in the steps.

2.1 The universal free energy and the universal spectrum

Setup and strategy

Work first at zero angular potential, $\beta_L = \beta_R = \beta$, and write

$$Z(\beta) = \sum_{\text{states}} e^{-\beta E}, \qquad E = \Delta - \frac{c}{12}, \qquad Z(\beta) = Z(\beta'), \quad \beta' \equiv \frac{4\pi^2}{\beta}. \tag{24}$$

Split the spectrum into three ranges [10, eq. (1.9)]:

$$\underbrace{-\frac{c}{12} \le E \le \varepsilon}_{\text{light}}, \qquad \underbrace{\varepsilon < E < \frac{c}{12}}_{\text{medium}}, \qquad \underbrace{E \ge \frac{c}{12}}_{\text{heavy}}, \tag{25}$$

where $\varepsilon > 0$ is a small parameter sent to zero at the end (it may be sent to zero with c, e.g. $\varepsilon \sim e^{-\alpha\sqrt{c}}$, at the cost of $o(c)$ errors). In terms of the total dimension: light is $\Delta \lesssim c/12$, medium is $c/12 < \Delta < c/6$, heavy is $\Delta \ge c/6$. The divisions are not arbitrary:

- $E = -c/12$ ($\Delta = 0$) is the vacuum;
- $E = 0$ ($\Delta = c/12$) is the BTZ threshold $M = 0$;
- $E = c/12$ ($\Delta = c/6$) is the Hawking–Page point, and by (37) (lecture 3) it is exactly where the Cardy saddle crosses the self-dual temperature.

Write $L = \{E \le \varepsilon\}$ and $H = \{E > \varepsilon\}$ for the light and non-light parts, and

$$Z[L] = \sum_L e^{-\beta E}, \quad Z[H] = \sum_H e^{-\beta E}, \qquad Z'[L] = \sum_L e^{-\beta' E}, \quad Z'[H] = \sum_H e^{-\beta' E}, \tag{26}$$

so that modular invariance reads

$$Z[L] + Z[H] = Z'[L] + Z'[H] \qquad \Longleftrightarrow \qquad Z[L] - Z'[L] = Z'[H] - Z[H]. \tag{27}$$

The strategy: the light spectrum is, by assumption, under control; the heavy spectrum is unknown. Modular invariance relates them, and (27) says that the *difference* of the heavy contributions in the two channels is fixed by the light spectrum. Combined with positivity — every ρ is non-negative, which is where unitarity enters — this is enough to bound the heavy contribution outright.

Bounding the heavy states by the light ones

Take $\beta > 2\pi$, so that $\beta' < 2\pi < \beta$ and $\beta' - \beta < 0$. Every state in H has $E > \varepsilon$, so

$$e^{-\beta E} = e^{(\beta'-\beta)E}\, e^{-\beta' E} \;\le\; e^{(\beta'-\beta)\varepsilon}\, e^{-\beta' E}, \tag{28}$$

term by term, because $\beta' - \beta < 0$ and $E > \varepsilon$. Summing over H,

$$Z[H] \;\le\; e^{(\beta'-\beta)\varepsilon}\, Z'[H]. \tag{29}$$

Rearranging and using (27),

$$Z'[H]\big(1 - e^{(\beta'-\beta)\varepsilon}\big) \;\le\; Z'[H] - Z[H] \;=\; Z[L] - Z'[L] \;\le\; Z[L], \tag{30}$$

the last step using $Z'[L] \geq 0$. Feeding this back into (29) gives the central inequality

$$\boxed{Z[H] \;\leq\; \frac{e^{(\beta'-\beta)\varepsilon}}{1-e^{(\beta'-\beta)\varepsilon}}\, Z[L], \qquad \beta > 2\pi.} \tag{31}$$

Two features. First, the prefactor is a pure number, independent of the theory: it depends only on β and ε. Second, it diverges as $\beta \to 2\pi$, where $\beta' \to \beta$. The bound is useless at the self-dual temperature and degrades continuously as one approaches it; everything below inherits this, so results hold away from $\beta = 2\pi$, with an error that is not uniform in β.

Sandwiching, and using $Z \geq Z[L]$,

$$\log Z[L] \;\leq\; \log Z(\beta) \;\leq\; \log Z[L] - \log\bigl(1 - e^{(\beta'-\beta)\varepsilon}\bigr), \qquad \beta > 2\pi, \tag{32}$$

and by modular invariance the mirror statement holds for $\beta < 2\pi$ with $Z'[L]$. In a family of theories with $c \to \infty$ the correction is $\mathcal{O}(c^0)$ while $\log Z = \mathcal{O}(c)$, so

$$\log Z(\beta) = \begin{cases} \log Z[L] + \mathcal{O}(1), & \beta > 2\pi, \\ \log Z'[L] + \mathcal{O}(1), & \beta < 2\pi. \end{cases} \tag{33}$$

The free energy of any modular-invariant CFT_2 at large c is determined, to leading order, by its light spectrum alone. No assumption about that spectrum has been used yet.

Sparseness as a necessary and sufficient condition

Equation (33) becomes a *universal* statement precisely when $Z[L]$ is dominated by the vacuum. Since the vacuum contributes $e^{\beta c/12}$,

$$Z[L] = e^{\beta c/12}\Bigl[1 + \sum_{0<\Delta\leq\frac{c}{12}+\varepsilon} \rho(\Delta)\, e^{-\beta\Delta}\Bigr], \tag{34}$$

and vacuum dominance to leading order in c is the statement

$$\log\Bigl(1 + \sum_{0<\Delta\leq\frac{c}{12}+\varepsilon} \rho(\Delta) e^{-\beta\Delta}\Bigr) = \mathcal{O}(1) \qquad \text{for all } \beta > 2\pi. \tag{35}$$

The sum is largest as $\beta \to 2\pi$ from above, so (35) holds for all $\beta > 2\pi$ if and only if it holds in that limit, i.e. if and only if

$$\rho(\Delta) \;\lesssim\; e^{2\pi\Delta} \qquad \Bigl(\Delta \leq \frac{c}{12}\Bigr), \tag{36}$$

which is the sparseness condition (17) [10, eq. (2.13)–(2.14)]. A theory violating it has a non-universal free energy; a theory satisfying it has

$$\boxed{\log Z(\beta) = \frac{c}{12}\,\max\Bigl(\beta,\; \frac{4\pi^2}{\beta}\Bigr) + \mathcal{O}(c^0).} \tag{37}$$

The two branches cross at $\beta = 2\pi$, where the free energy has a first-order phase transition. In gravity this is Hawking–Page; here it follows from modular invariance and a bound on the light spectrum, with no gravitational input.

The statement runs in both directions. *Gravity* $\Rightarrow$ *sparseness*: if a CFT has a semiclassical AdS_3 dual its free energy must be (37), hence its light spectrum must be sparse — sparseness is a necessary condition for holography, not an optional extra. *Sparseness* $\Rightarrow$ *gravity*: conversely, sparseness plus modular invariance produces (37) with no further input. At the level of the leading free energy, "holographic" has an exact characterisation.

From the free energy to the spectrum

One would like to Legendre transform (37) to obtain $S(E)$. This requires care. The thermodynamic energy and entropy following from (37) are

$$E(\beta) = -\partial_\beta \log Z = \begin{cases} -\dfrac{c}{12} + \mathcal{O}(1), & \beta > 2\pi, \\ \dfrac{\pi^2 c}{3\beta^2} + \mathcal{O}(1), & \beta < 2\pi, \end{cases} \tag{38}$$

$$S(\beta) = (1 - \beta\partial_\beta) \log Z = \begin{cases} \mathcal{O}(1), & \beta > 2\pi, \\ \dfrac{2\pi^2 c}{3\beta} + \mathcal{O}(1), & \beta < 2\pi, \end{cases} \tag{39}$$

[checked]. At $\beta = 2\pi$ the energy jumps from $-c/12$ to $+c/12$: the canonical ensemble never samples the window $-c/12 < E < c/12$ at large c. For finite c the curve $E(\beta)$ is of course continuous; what (38) means is that an $\mathcal{O}(1)$ ambiguity in $\log Z$ near $\beta = 2\pi$ — exactly the ambiguity left by (32) — produces an $\mathcal{O}(c)$ ambiguity in E. This is the degeneration of (31) seen from the microcanonical side.

The Legendre transform can therefore only be trusted where the saddle is safely on one side of the transition, $E > c/12$, and there

$$\boxed{S(E) \;\sim\; 2\pi\sqrt{\frac{c}{3}E} \qquad \text{for} \qquad E > \frac{c}{12}, \quad \text{i.e. } \Delta > \frac{c}{6}.} \tag{40}$$

This holds not merely for the thermodynamic entropy but for the microscopic density of states, $\rho(E) \approx e^{S(E)}$, once one smears over an energy window $\delta E \sim c^\alpha$ with $\frac{1}{2} < \alpha < 1$ — large enough to contain many states, small enough not to blur the exponent [10, App. A].

Compare with §1.2 (lecture 3): the self-consistency of the Cardy saddle required $E_L, E_R > c/24$, i.e. $E > c/12$ in the non-rotating case. The theorem says that this naive condition is the correct one, provided the light spectrum is sparse. The much stronger requirement $E \gg c$ of (40) (lecture 3) was an artefact of not knowing what to assume about the light states.

Non-zero angular potential

For black holes one needs $\beta_L \neq \beta_R$. The argument above does not immediately generalise, because the split into light and heavy is now two-dimensional and the region where $Z[H]$ can be bounded by $Z[L]$ is only part of the (β_L, β_R) plane. HKS proceed iteratively [10, §3]. In the quadrants $\beta_L, \beta_R > 2\pi$ and $\beta_L, \beta_R < 2\pi$ the non-rotating bound (17) already suffices: for $\beta_{L,R} > 2\pi$ one has $\rho(E_L, E_R) \leq \rho(E_L + E_R) \lesssim e^{\frac{\pi c}{6} + 2\pi(E_L + E_R)}$, so the exponent in $\sum \rho\, e^{-\beta_L E_L - \beta_R E_R}$ obeys

$$\frac{\pi c}{6} + 2\pi(E_L + E_R) - \beta_L E_L - \beta_R E_R \;\leq\; \frac{c}{24}\big(\beta_L + \beta_R\big), \tag{41}$$

since for $\beta_{L,R} > 2\pi$ the left side decreases in $E_{L,R}$ and is therefore largest at $E_L = E_R = -c/24$ — the vacuum, which saturates it [checked]. Every other state is exponentially subleading [10, eq. (3.3)–(3.6)]. The free energy so obtained constrains the spectrum; those constraints extend the region in which the free energy can be shown to be universal; and so on. After three iterations the only region left is a thin sliver around the self-dual locus $\beta_L \beta_R = 4\pi^2$, which HKS conjecture

further iterations would remove. Assuming in addition the mixed sparseness condition (18), the result is

$$\boxed{\log Z(\beta_L,\beta_R) = \frac{c}{24}\max\Big(\beta_L+\beta_R,\ \frac{4\pi^2}{\beta_L}+\frac{4\pi^2}{\beta_R}\Big)+\mathcal{O}(c^0),} \tag{42}$$

away from $\beta_L\beta_R = 4\pi^2$, with the corresponding spectrum

$$S(E_L,E_R)\ \sim\ 2\pi\sqrt{\frac{c\,E_L}{6}}+2\pi\sqrt{\frac{c\,E_R}{6}} \qquad \text{for} \qquad E_LE_R > \Big(\frac{c}{24}\Big)^2. \tag{43}$$

The rigorously established range is $E_L, E_R > c/24$, both chiralities above the transition separately; the extension to the hyperbola $E_LE_R > (c/24)^2$ is the conjectural part [10, eq. (3.10),(3.30)–(3.32)]. Setting $\beta_L = \beta_R$ in (42) returns (37), and $E_L = E_R$ in (43) returns (40).

2.2 The window $c/12 < \Delta < c/6$

Between the BTZ threshold and the Hawking–Page point lies a range the canonical ensemble never explores. HKS call it the "enigmatic" range. The entropy there is *not* universal, but it is bounded.

Upper bound. Positivity gives $Z(\beta) > \rho(E)e^{-\beta E}$ for any single energy and any β. Evaluate at the self-dual point $\beta = 2\pi$, where $\log Z = \pi c/6$ by (37):

$$S(E)\ \lesssim\ \frac{\pi c}{6}+2\pi E \qquad\Longleftrightarrow\qquad \rho(\Delta)\ \lesssim\ e^{2\pi\Delta} \tag{44}$$

[checked] (using $E = \Delta - c/12$). The bound has the same form as the sparseness condition (17): sparseness is *imposed* below $\Delta = c/12$ and *derived* between $c/12$ and $c/6$. With an angular potential the same argument gives $\rho(\Delta_L,\Delta_R) \lesssim e^{4\pi\sqrt{\Delta_L\Delta_R}}$ [10, eq. (3.31)], again the form of (18).

Lower bound. There is no rigorous lower bound: HKS note that their results are compatible with there being no primaries at all in this range [10, §2.4]. Heuristically, though, rewrite the heavy contribution using modular invariance,

$$Z[H] = Z'[L] + \big(Z'[H] - Z[L]\big), \tag{45}$$

and note that $Z'[L]$ contains the vacuum term $e^{\beta' c/12}$. Unless the heavy spectrum conspires to cancel it, one expects a corresponding contribution to the density of states, giving

$$2\pi\sqrt{\frac{c}{3}E}\ \lesssim\ S(E)\ \lesssim\ \frac{\pi c}{6}+2\pi E \qquad \Big(0 < E < \frac{c}{12}\Big). \tag{46}$$

Both ends are attained: symmetric orbifolds saturate the upper bound (§2.3), and a hypothetical pure-gravity CFT would saturate the lower one.

Why it does not affect the free energy. A state obeying (44) contributes to $Z(\beta)$ at most

$$\rho(E)\,e^{-\beta E}\ \lesssim\ \exp\Big[\frac{\pi c}{6}+(2\pi-\beta)E\Big], \tag{47}$$

which for $\beta > 2\pi$ is maximised at $E \to 0$, where it equals $e^{\pi c/6}$ — smaller than the vacuum contribution $e^{\beta c/12} > e^{\pi c/6}$. The medium states therefore never dominate, and (37) is unaffected by our ignorance of them [checked].

The bulk counterpart. In gravity this window is $0 < r_+ < \ell$, by (4). There the BTZ black hole exists and has positive specific heat — $M = r_+^2/8G\ell^2$ and $T = r_+/2\pi\ell^2$ give $M \propto T^2$, so $\mathrm{d}M/\mathrm{d}T > 0$ everywhere [checked] — but its free energy exceeds that of thermal AdS_3, so it is a subdominant saddle. (This is what "thermodynamically unstable" is usually taken to mean in this context; it is not the negative-specific-heat instability of a Schwarzschild black hole, which BTZ never has.) In AdS_3 *with matter* — for instance in the actual D1–D5 bulk — further saddles can exist in this range. The clearest example is the family of [19]: supersymmetric multi-centred configurations in which a black hole sheds part of its charge into a condensate outside the horizon, with more entropy than the black hole alone over a range of charges, while still obeying (44). They never dominate the canonical ensemble, which is why the leading free energy is universal even though the spectrum in this range is not.

2.3 Symmetric orbifolds

The generating function (58) (lecture 3) makes $\mathrm{Sym}^N(\mathcal{C})$ explicit enough to test all of the above, and HKS prove [10, §5, App. C] that for *any* seed:

(a) the free energy is (42) for *all* $\beta_L, \beta_R > 0$ — stronger than the general theorem, since it covers the sliver as well;

(b) the density of states saturates the sparseness bound in the region $E_L E_R < (c/24)^2$, as in (23), and equals the Cardy value (43) above that hyperbola. The two expressions agree on the hyperbola itself: at $\Delta_L = \Delta_R = c/12$ both give $\pi c/3$ [checked].

The mechanism is the long string of §2.4 (lecture 3). For given (E_L, E_R) the dominant configuration consists of m short strings together with one long string of length $N - m$, and the entropy is obtained by maximising over m. At large energy the long string takes up everything and one gets Cardy; at small energy the optimum is a partially long string and one gets the saturating answer.

Two consequences we have been building towards.

The Strominger–Vafa regime problem is solved. For $\mathrm{Sym}^N(M_4)$ — and, by the necessary-and-sufficient logic of §2.1, for the D1–D5 theory at any point of $\mathcal{M}$ whose light spectrum is sparse — the Cardy formula applies throughout $E_L E_R > (c/24)^2$. It is not restricted to $N_p \gg Q_1 Q_5$.

Maximal density is not a virtue. Saturating (17) is the boundary case: enough for the leading free energy, and exactly Hagedorn. A theory that saturates it is consistent with $3d$ gravity at leading order in $1/c$ but has a string at the AdS scale.

2.4 Back to black holes

Collect the pieces. Let $\mathcal{T}$ be a unitary, modular-invariant, compact CFT_2 with $c = \bar{c} = 3\ell/2G \gg 1$ and a sparse light spectrum, (17) and (18). Then for every (E_L, E_R) with $E_L E_R > (c/24)^2$,

$$S_{\mathrm{micro}}(E_L, E_R) = 2\pi\sqrt{\frac{cE_L}{6}} + 2\pi\sqrt{\frac{cE_R}{6}} = \frac{\pi(r_+ + r_-)}{4G} + \frac{\pi(r_+ - r_-)}{4G} = \frac{2\pi r_+}{4G} = \frac{A}{4G}, \tag{48}$$

[checked symbolically]. What has changed since §1.2 (lecture 3) is the validity condition. Using (4) and $c = 3\ell/2G$,

$$E_L E_R = \frac{(r_+^2 - r_-^2)^2}{(16G\ell)^2}, \qquad \left(\frac{c}{24}\right)^2 = \left(\frac{\ell}{16G}\right)^2, \tag{49}$$

so

$$\boxed{E_L E_R > \left(\frac{c}{24}\right)^2 \qquad \Longleftrightarrow \qquad r_+^2 - r_-^2 > \ell^2} \tag{50}$$

[checked symbolically], which is exactly the condition that the black hole lies *above* the Hawking–Page transition: from lecture 2, §1.6 the rotating transition locus is $r_+^2 - r_-^2 = \ell^2$, re-derived from the action in §3.4. Likewise the rigorously established range $E_L, E_R > c/24$ becomes $r_+ + r_- > \ell$ and $r_+ - r_- > \ell$ [checked].

The statement is therefore:

In any large-c, unitary, modular-invariant CFT_2 *with a sparse light spectrum, the microcanonical entropy equals A/4G for every BTZ black hole that dominates the canonical ensemble.*

The two conditions coincide for a reason. The Cardy saddle sits at $\beta^*_{L,R} = \pi\sqrt{c/6E_{L,R}}$; the free energy has its phase transition at $\beta_L\beta_R = 4\pi^2$; and $\beta^*_L\beta^*_R < 4\pi^2$ is precisely $E_L E_R > (c/24)^2$. The same self-dual point governs both. Where the derivation fails, $r_+^2 - r_-^2 < \ell^2$, the black hole is not the dominant saddle, so there is nothing there for the entropy to match.

The difference from §1.2 (lecture 3) is the following. There, the central charge came from the bulk and the density of states was assumed to be Cardy's; the assumption was uncontrolled precisely in the regime of interest. Here the density of states is *derived*, from modular invariance and one hypothesis about light states, and the regime in which the derivation works coincides with the regime in which the object being described exists as a dominant saddle. The price is that the hypothesis is not free: §1.4 shows that most large-c theories fail it, and §2.1 shows that any theory with a gravity dual must satisfy it.

Three limitations remain.

- *Below Hawking–Page the answer is genuinely non-universal*, (46). Two holographic CFT's can have different densities of states there, matching the fact that two bulk theories can have different subdominant saddles.
- *Nothing here fixes the* $\mathcal{O}(c^0)$ *terms.* Those are the one-loop determinants around each saddle and the sum over the $\mathrm{SL}(2,\mathbb{Z})$ family of geometries, where the Maloney–Witten problem lives.
- *Nothing here says the bulk is Einstein gravity.* Sparseness is compatible with a tensionless string. Distinguishing the two needs the gap (15) and strictly sub-Hagedorn growth (21), neither of which is visible in the leading free energy.

Exercise 2. Locate in the (r_+, r_-) plane the three regions: (i) the rigorous range $r_+ \pm r_- > \ell$; (ii) the conjectural extension $r_+^2 - r_-^2 > \ell^2$; (iii) the enigmatic window $r_+^2 - r_-^2 < \ell^2$. Show that (i) and (ii) differ only near extremality, $r_+ \approx r_-$, so that the entropy of near-extremal BTZ black holes is covered only by the conjectural part of the analysis. This is one reason near-extremal AdS_3 black holes, and their AdS_2 throats, need separate treatment.

Exercise 3. Show that (42) is the same function as $\max\big(\log Z_{\mathrm{therm}}, \log Z_{\mathrm{BTZ}}\big)$ with $\log Z_{\mathrm{therm}} = \frac{c}{24}(\beta_L + \beta_R)$ and $\log Z_{\mathrm{BTZ}} = \frac{\pi^2 c}{6}\big(\beta_L^{-1} + \beta_R^{-1}\big)$. We compute both from the gravitational action in §3.

3 The gravitational path integral at tree level

We now compute, from the gravitational action, the quantity that (42) predicts. The framework is Gibbons–Hawking [20]: the Euclidean path integral over metrics with prescribed boundary data

computes a thermodynamic partition function,

$$Z(\beta_L,\beta_R)=\int_{\partial\mathcal{M}=T^2}\mathcal{D}g\; e^{-I_E[g]}\;\approx\;\sum_{\text{saddles}} e^{-I_E^{\text{cl}}}\quad\text{at large } c, \tag{51}$$

with the boundary torus fixed by (β_L,β_R).

3.1 Boundary terms: the variational principle and the divergences

Write the Euclidean bulk action as

$$I_E^{\text{bulk}}=-\frac{1}{16\pi G}\int_{\mathcal{M}}\mathrm{d}^3x\,\sqrt{g}\,\Big(R+\frac{2}{\ell^2}\Big). \tag{52}$$

The variational principle. Varying (52),

$$\delta\big[\sqrt{g}(R+2/\ell^2)\big]=\sqrt{g}\,\big(G_{\mu\nu}-\tfrac{1}{\ell^2}g_{\mu\nu}\big)\delta g^{\mu\nu}+\sqrt{g}\,\nabla_\mu v^\mu,\qquad v^\mu=g^{\mu\nu}g^{\alpha\beta}\big(\nabla_\beta\delta g_{\nu\alpha}-\nabla_\nu\delta g_{\alpha\beta}\big). \tag{53}$$

The first term gives the equations of motion. The second is a total derivative and becomes a boundary integral of $n_\mu v^\mu$, which contains derivatives of $\delta g_{\mu\nu}$ *normal* to the boundary; explicitly, with the induced metric held fixed,

$$n_\mu v^\mu\big|_{\delta h_{ij}=0}=-2\,\delta K. \tag{54}$$

Setting $\delta h_{ij}=0$ — the Dirichlet problem, which is the one appropriate to a CFT on a fixed background metric — therefore does *not* make the variation vanish: one would have to fix $\partial_n\delta h_{ij}$ as well, which is not a well-posed boundary condition for a second-order equation. The cure is the Gibbons–Hawking–York term [21, 20]

$$\boxed{I_E^{\text{GH}}=-\frac{1}{8\pi G}\oint_{\partial\mathcal{M}}\mathrm{d}^2x\,\sqrt{h}\,K,\qquad K_{ij}=\tfrac{1}{2}\pounds_n h_{ij},\quad K=h^{ij}K_{ij},} \tag{55}$$

with n^μ the outward unit normal. Its variation cancels the δK, leaving

$$\delta\big(I_E^{\text{bulk}}+I_E^{\text{GH}}\big)=(\text{eom})\;-\;\frac{1}{16\pi G}\oint\mathrm{d}^2x\,\sqrt{h}\,\big(K^{ij}-Kh^{ij}\big)\delta h_{ij}, \tag{56}$$

which vanishes on shell for $\delta h_{ij}=0$. Without (55) no classical solution extremises the action and e^{-I_E} has no saddle-point interpretation.

The divergences. With (55) included the action is still infinite, because $\partial\mathcal{M}$ is at infinity. Regulate by placing the boundary at large finite radius. For thermal AdS_3 one finds (§3.3) $I_E^{\text{bulk}}\sim+\beta r_c^2/4G\ell^2$ and $I_E^{\text{GH}}\sim-\beta r_c^2/2G\ell^2$: both divergent, and their sum still divergent. The divergence is, however, a *local* functional of the boundary data, proportional to $\oint\sqrt{h}$, the volume of the regulating surface. In three bulk dimensions the only covariant boundary counterterms available are $\oint\sqrt{h}$ and $\oint\sqrt{h}\,\mathcal{R}[h]$, and the second is topological in two dimensions, so one term suffices [23, 24, 25]:

$$\boxed{I_E^{\text{ct}}=+\frac{1}{8\pi G\ell}\oint_{\partial\mathcal{M}}\mathrm{d}^2x\,\sqrt{h}.} \tag{57}$$

Adding it does not spoil (56): built from h_{ij} alone, it contributes an extra $\frac{1}{16\pi G\ell}\sqrt{h}\,h^{ij}\delta h_{ij}$, i.e. a finite shift of the stress tensor, and nothing involving normal derivatives. The renormalised action is

$$I_E = I_E^{\text{bulk}} + I_E^{\text{GH}} + I_E^{\text{ct}}. \tag{58}$$

This is *holographic renormalisation*: the counterterm is the bulk avatar of the local counterterms that render $\log Z$ of a CFT on a curved background finite, and the residual freedom in choosing it is the usual CFT scheme dependence. Three things become finite at once — the on-shell action, the boundary stress tensor, and the conserved charges — and we do the stress tensor first, because it carries the physical information.

Remark 2. An older prescription is *background subtraction*: regulate at $r = r_c$ and subtract the action of a reference geometry (here thermal AdS_3) whose period is adjusted so that the two induce the same boundary metric at $r = r_c$. This also gives a finite answer and the same free-energy *differences*, hence the same Hawking–Page temperature, but a different absolute normalisation: $M_{AdS_3} = 0$ rather than $-1/8G$. The counterterm scheme is the one that agrees with the CFT on the cylinder, because it assigns the vacuum its Casimir energy $-c/12\ell$, and we use it throughout.

3.2 The boundary stress tensor, mass and angular momentum

Varying the renormalised action with respect to the boundary metric defines the quasilocal (Brown–York) stress tensor [22, 23]. From (56) together with (57),

$$\boxed{T_{ij} = -\frac{1}{8\pi G}\Big(K_{ij} - K\,h_{ij} + \frac{1}{\ell}h_{ij}\Big),} \tag{59}$$

where n^μ points *towards* the boundary; the overall sign is fixed by matching to the CFT convention of lecture 2, §2.2, and is verified below.

The natural place to evaluate (59) is the Fefferman–Graham gauge of lecture 2, §2, in which the whole solution space is already parametrised. Recall the Bañados family:

$$\mathrm{d}s^2 = \ell^2\Big[\frac{\mathrm{d}\rho^2}{\rho^2} + \mathcal{L}(x^+)(\mathrm{d}x^+)^2 + \bar{\mathcal{L}}(x^-)(\mathrm{d}x^-)^2 - \Big(\frac{1}{\rho^2} + \rho^2\mathcal{L}\bar{\mathcal{L}}\Big)\mathrm{d}x^+\mathrm{d}x^-\Big], \tag{60}$$

with the conformal boundary at $\rho \to 0$. Then $g_{\rho\rho} = \ell^2/\rho^2$, the unit normal pointing towards the boundary is $n^\mu\partial_\mu = -\frac{\rho}{\ell}\partial_\rho$, and

$$K_{ij} = \tfrac{1}{2}\pounds_n h_{ij} = -\frac{\rho}{2\ell}\,\partial_\rho h_{ij}, \qquad K = h^{ij}K_{ij} = \frac{2}{\ell}\big(1 + \mathcal{O}(\rho^4)\big) \tag{61}$$

[checked symbolically]. The leading behaviour $K \to d/\ell$ with $d = 2$ is what the counterterm cancels: $K_{ij} \to h_{ij}/\ell$ at leading order, so $K_{ij} - Kh_{ij} \to -h_{ij}/\ell$, which $+h_{ij}/\ell$ removes. Carrying this out,

$$\boxed{T_{++} = \frac{\ell}{8\pi G}\,\mathcal{L} = \frac{c}{12\pi}\,\mathcal{L}, \qquad T_{--} = \frac{\ell}{8\pi G}\,\bar{\mathcal{L}} = \frac{c}{12\pi}\,\bar{\mathcal{L}}, \qquad T_{+-} = \mathcal{O}(\rho^2) \to 0,} \tag{62}$$

[checked symbolically] as $\rho \to 0$. Three comments.

- *Finiteness.* Every divergent piece cancelled: the $1/\rho^2$ in h_{+-} appears in both K_{ij} and Kh_{ij} and is removed by the counterterm. Without (57) the stress tensor would diverge like $1/\rho^2$.

- *Tracelessness.* $T_{+-} = 0$ says $T^i{}_i = 0$: no conformal anomaly, because the boundary metric $\gamma = -\ell^2 \mathrm{d}x^+ \mathrm{d}x^-$ is flat. For a curved boundary metric one finds instead $\langle T^i{}_i \rangle \propto c\,\mathcal{R}[\gamma]$, the holographic Weyl anomaly [26, 25].
- *Chirality.* T_{++} depends only on x^+ and T_{--} only on x^-: the boundary stress tensor is a pair of chiral currents, as it must be in a CFT_2. The Bañados functions $\mathcal{L}, \bar{\mathcal{L}}$ are the holomorphic and antiholomorphic stress tensors up to normalisation.

The conserved charges

The boundary metric read off from (60) at $\rho \to 0$ is $h_{ij} \to \gamma_{ij}/\rho^2$ with

$$\gamma = -\ell^2 \, \mathrm{d}x^+ \mathrm{d}x^- = -\mathrm{d}t^2 + \ell^2 \mathrm{d}\varphi^2, \tag{63}$$

a flat cylinder of circumference $2\pi\ell$. Since $d = 2$, the stress tensor (62) has conformal weight zero and needs no rescaling. For a Killing vector ξ of γ the conserved charge is

$$Q[\xi] = \oint \mathrm{d}\varphi \, \sqrt{\sigma} \, u^i \, T_{ij} \, \xi^j, \tag{64}$$

with $u = \partial_t$ the unit normal to the constant-t circle and $\sqrt{\sigma} = \ell$ its induced measure. Using $\partial_t = \frac{1}{\ell}(\partial_+ + \partial_-)$ and $\partial_\varphi = \partial_+ - \partial_-$,

$$M \equiv Q[\partial_t] = \frac{2\pi}{\ell}\big(T_{++} + T_{--}\big), \qquad J \equiv Q[\partial_\varphi] = 2\pi\big(T_{++} - T_{--}\big), \tag{65}$$

[checked symbolically], i.e., comparing with the Conventions,

$$\boxed{E_L = \oint_0^{2\pi} \mathrm{d}\varphi \, T_{++} = \frac{c}{6}\mathcal{L}, \qquad E_R = \oint_0^{2\pi} \mathrm{d}\varphi \, T_{--} = \frac{c}{6}\bar{\mathcal{L}}} \tag{66}$$

[checked symbolically]: the conserved charges are the zero modes of the chiral stress tensors, which is the statement $E_L = L_0 - c/24$, $E_R = \bar{L}_0 - c/24$. This turns the identification $L_0 = \frac{c}{6}\mathcal{L}$, a normalisation choice in lecture 2, §2.2, into a computation.

Now evaluate, using $\mathcal{L} = 2G(M + J/\ell) = (r_+ + r_-)^2/4\ell^2$ and $\bar{\mathcal{L}} = 2G(M - J/\ell) = (r_+ - r_-)^2/4\ell^2$ from lecture 2, eq. (36).

Global AdS_3. Here $\mathcal{L} = \bar{\mathcal{L}} = -\frac{1}{4}$, so

$$\boxed{M_{\mathrm{AdS}_3} = -\frac{1}{8G} = -\frac{c}{12\,\ell}, \qquad J_{\mathrm{AdS}_3} = 0} \tag{67}$$

[checked symbolically]. This checks the counterterm scheme rather than assuming it: the Casimir energy of a CFT_2 on a circle of circumference L is $-\pi c/6L$, and with $L = 2\pi\ell$ that is $-c/12\ell$ [checked]. Equivalently $E_L = E_R = -c/24$, i.e. $L_0 = \bar{L}_0 = 0$: global AdS_3 is the CFT vacuum.

Rotating BTZ. Inserting $\mathcal{L}, \bar{\mathcal{L}}$,

$$\boxed{M = \frac{r_+^2 + r_-^2}{8G\ell^2}, \qquad J = \frac{r_+ r_-}{4G\ell}} \tag{68}$$

[checked symbolically], reproducing (2), with $M\ell \geq |J|$ and equality at extremality. The $M = J = 0$ geometry ($\mathcal{L} = \bar{\mathcal{L}} = 0$) has $M = 0$, not $-1/8G$: the counterterm scheme distinguishes it from global AdS_3, and the difference $c/12\ell$ is the Casimir energy — the shift between the Ramond and Neveu–Schwarz ground states used in §1.2 (lecture 3).

3.3 On-shell actions

All the geometries we need have the form

$$\mathrm{d}s_E^2 = f(r)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\big(\mathrm{d}\varphi + N^\varphi(r)\,\mathrm{d}\tau\big)^2, \qquad \varphi \simeq \varphi + 2\pi, \tag{69}$$

obtained from the Lorentzian solutions by $t \to -i\tau$ (and, for the rotating case, $r_- \to i\mathfrak{r}$, so that the Euclidean section is real; we continue back at the end). For all of them

$$\sqrt{g} = r, \tag{70}$$

[checked symbolically], independently of N^φ: the shear in the (τ, φ) identification does not change the area of the torus. On a surface of constant r,

$$\sqrt{h} = r\sqrt{f}, \qquad K = \frac{1}{r}\partial_r\big(r\sqrt{f}\big), \qquad \sqrt{h}\,K = f + \frac{rf'}{2}, \tag{71}$$

[checked symbolically]. Every vacuum solution has $R = 6\Lambda = -6/\ell^2$, so the bulk integrand is a constant, $R + 2/\ell^2 = -4/\ell^2$, and

$$I_E^{\mathrm{bulk}} = \frac{1}{4\pi G\ell^2}\,\mathrm{Vol}(\mathcal{M}), \qquad \mathrm{Vol} = 2\pi\beta\int_{r_{\min}}^{r_c} r\,\mathrm{d}r = \pi\beta\big(r_c^2 - r_{\min}^2\big). \tag{72}$$

Altogether

$$I_E^{\mathrm{bulk}} = \frac{\beta\big(r_c^2 - r_{\min}^2\big)}{4G\ell^2}, \qquad I_E^{\mathrm{GH}} = -\frac{\beta}{4G}\Big(f + \frac{rf'}{2}\Big)\Big|_{r=r_c}, \qquad I_E^{\mathrm{ct}} = \frac{\beta}{4G\ell}\Big(r\sqrt{f}\Big)\Big|_{r=r_c}, \tag{73}$$

[checked], where r_c is the location of the regulating boundary and $r_{\min}$ the inner end of the r integration.

Rotating BTZ. Take

$$f = \frac{(r^2 - r_+^2)(r^2 - r_-^2)}{\ell^2 r^2}, \qquad N^\varphi = -\frac{r_+ r_-}{\ell r^2}, \qquad r \in [r_+, r_c], \qquad \beta = \frac{4\pi}{f'(r_+)} = \frac{2\pi\ell^2 r_+}{r_+^2 - r_-^2}, \tag{74}$$

the period fixed by smoothness at $r = r_+$, with the simultaneous identification $\varphi \to \varphi + \beta\Omega_H$ — real on the Euclidean section, where $r_- \to i\mathfrak{r}$ makes both f and $-iN^\varphi$ real [checked]. Then

$$I_E^{\mathrm{bulk}} + I_E^{\mathrm{GH}} + I_E^{\mathrm{ct}} = \frac{\beta}{4G\ell^2}\Big[-r_c^2 + r_-^2 + \sqrt{(r_c^2 - r_+^2)(r_c^2 - r_-^2)}\Big] \tag{75}$$

[checked symbolically] at finite r_c; the r_c^2 terms cancel and, expanding the square root,

$$\boxed{I_{\mathrm{BTZ}} = -\frac{\beta\,(r_+^2 - r_-^2)}{8G\ell^2} = -\frac{\pi r_+}{4G}} \tag{76}$$

[checked symbolically]. Setting $r_- = 0$ gives the static case, $I = -\beta r_+^2/8G\ell^2 = -\beta M$.

Two checks. First, $I = \beta F$ with F the Gibbs free energy, so (76) should equal $\beta(M - \Omega_H J) - S$. Using (68), $S = \pi r_+/2G$ and $\Omega_H = r_-/\ell r_+$,

$$\beta\big(M - \Omega_H J\big) - S = \frac{2\pi\ell^2 r_+}{r_+^2 - r_-^2}\cdot\frac{r_+^2 - r_-^2}{8G\ell^2} - \frac{\pi r_+}{2G} = \frac{\pi r_+}{4G} - \frac{\pi r_+}{2G} = -\frac{\pi r_+}{4G} \tag{77}$$

[checked symbolically], identically in (r_+, r_-). This ties together three independently computed objects: the Euclidean action, the counterterm-renormalised charges, and the horizon area. Second, in the dimensionless variables (5),

$$\boxed{\log Z_{\rm BTZ} = -I_{\rm BTZ} = \frac{c}{24}\Big(\frac{4\pi^2}{\beta_L} + \frac{4\pi^2}{\beta_R}\Big) = \frac{\pi^2 c}{6}\Big(\frac{1}{\beta_L} + \frac{1}{\beta_R}\Big)} \tag{78}$$

[checked symbolically].

Thermal AdS_3**.** Take

$$f = 1 + \frac{r^2}{\ell^2}, \qquad N^\varphi = 0, \qquad r \in [0, r_c], \tag{79}$$

with $\tau \simeq \tau + \beta$ for *arbitrary* β (there is no horizon to fix it) and, if an angular potential is wanted, the identification $(\tau, \varphi) \simeq (\tau + \beta, \varphi + \beta\Omega)$. Then

$$I_E^{\rm bulk} = \frac{\beta r_c^2}{4G\ell^2}, \qquad I_E^{\rm GH} = -\frac{\beta}{4G}\Big(1 + \frac{2r_c^2}{\ell^2}\Big), \qquad I_E^{\rm ct} = \frac{\beta r_c\sqrt{1 + r_c^2/\ell^2}}{4G\ell} = \frac{\beta r_c^2}{4G\ell^2} + \frac{\beta}{8G} + \mathcal{O}(r_c^{-2}), \tag{80}$$

[checked symbolically]. The r_c^2 divergences cancel between the three terms ($\frac{1}{4} - \frac{1}{2} + \frac{1}{4} = 0$), and

$$\boxed{I_{\mathrm{AdS}_3} = -\frac{\beta}{8G} \qquad \Longrightarrow \qquad \log Z_{\rm therm} = \frac{\beta}{8G} = \frac{c}{12}\hat{\beta} = \frac{c}{24}(\beta_L + \beta_R)} \tag{81}$$

[checked symbolically]. Note $I = \beta M_{\mathrm{AdS}_3}$: thermal AdS_3 has no horizon, so $S = 0$ and its free energy is just its (negative, Casimir) energy, independent of β at this order — and independent of Ω, as (81) shows, since the shear does not change the volume. Temperature dependence enters only at one loop, through the boundary-graviton determinant.

3.4 The sum over saddles

Two fillings of the same torus. The boundary is a torus with cycles φ and τ. A smooth Euclidean solution must fill it in, and the two saddles differ by which cycle becomes contractible in the interior. In thermal AdS_3, $g_{\varphi\varphi} = r^2 \to 0$ at $r = 0$ while $g_{\tau\tau} \to 1$: the *spatial* circle shrinks, smoothness at the origin is guaranteed by $\varphi \simeq \varphi + 2\pi$, and β is left free — which is why (81) holds for arbitrary β. In Euclidean BTZ, $g_{\tau\tau} = f \to 0$ at $r = r_+$: the *thermal* circle shrinks, and smoothness there fixes β. The two geometries are the same solid torus with the roles of the two cycles exchanged, which is exactly the operation $\tau \to -1/\tau$ of (7). This is the geometric reason the two saddles are exchanged at the self-dual point $\beta_L = \beta_R = 2\pi$, and the reason the complete list of fillings is labelled by $\mathrm{SL}(2, \mathbb{Z})$.

Both geometries are smooth Euclidean solutions with the same boundary torus, so both contribute:

$$Z_{\rm grav}(\beta_L, \beta_R) \approx e^{-I_{\mathrm{AdS}_3}} + e^{-I_{\rm BTZ}} + \cdots, \tag{82}$$

and at large c the smaller action wins:

$$\boxed{\log Z_{\rm grav}(\beta_L, \beta_R) = \frac{c}{24}\max\Big(\beta_L + \beta_R,\ \frac{4\pi^2}{\beta_L} + \frac{4\pi^2}{\beta_R}\Big) + \mathcal{O}(c^0).} \tag{83}$$

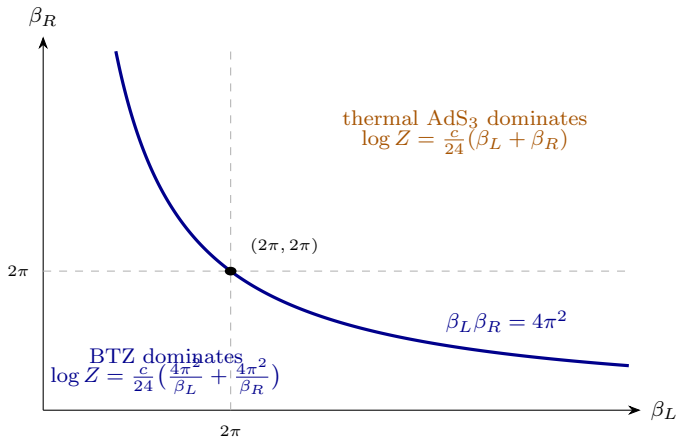


Figure 1: The Hawking–Page phase diagram in the (β_L, β_R) plane. The blue curve is the self-dual locus $\beta_L\beta_R = 4\pi^2$, equivalently $r_+^2 - r_-^2 = \ell^2$. Above it thermal AdS_3 has the smaller action; below it the BTZ black hole does. The CFT derivation of the same diagram, from modular invariance and sparseness [10], leaves a thin sliver around the blue curve in which universality is conjectured but not proved.

This is the same function as (42). The gravitational tree-level free energy and the universal free energy of a large-c CFT_2 with a sparse light spectrum agree identically, as functions of two variables, including the location of the phase transition.

Differentiating the BTZ branch closes the loop back to §1.2 (lecture 3). With $E_{L,R} = -\partial_{\beta_{L,R}} \log Z = \pi^2 c/6\beta_{L,R}^2$,

$$S = \left(1 - \beta_L\partial_{\beta_L} - \beta_R\partial_{\beta_R}\right)\log Z_{\text{BTZ}} = \frac{\pi^2 c}{3}\left(\frac{1}{\beta_L} + \frac{1}{\beta_R}\right) = \frac{\pi(r_+ + r_-)}{4G} + \frac{\pi(r_+ - r_-)}{4G} = \frac{A}{4G} \tag{84}$$

[checked symbolically]. The tree-level gravitational path integral reproduces the Bekenstein–Hawking entropy, as it must; the same function (42) reproduces it on the CFT side.

Explicitly, in the non-rotating case with $x \equiv 2\pi\ell T = 2\pi/\hat{\beta}$,

$$8G\,F_{\text{AdS}_3} = -1, \qquad 8G\,F_{\text{BTZ}} = -x^2, \tag{85}$$

[checked], crossing at $x = 1$, i.e. $\hat{\beta} = 2\pi$, $r_+ = \ell$. With rotation the transition locus is $\beta_L\beta_R = 4\pi^2$, i.e.

$$r_+^2 - r_-^2 = \ell^2, \tag{86}$$

[checked symbolically] — the condition (50) that appeared in §2.4 as the range of validity of the Cardy formula.

What the agreement does and does not establish.

- *Gravity determines the* CFT*'s light spectrum.* Given the bulk, (83) follows from two classical solutions. Any CFT dual must reproduce it, and by §2.1 that forces sparseness. Sparseness is therefore a prediction about holographic CFT's rather than an assumption bolted on.
- *The* CFT *determines the leading bulk thermodynamics.* Conversely, sparseness plus modular invariance gives (42) with no bulk input.

- *The $\mathcal{O}(c^0)$ terms are untouched.* These are the one-loop determinants around each saddle. Around thermal AdS_3 the answer is the vacuum Virasoro character $\prod_{n\geq 2}|1-q^n|^{-2}$ [27, 28]: a gas of boundary gravitons, whose $\log\rho \sim \Delta^{1/2}$ spectrum appeared in §1.3. Beyond that, the two saddles above are only the first two members of an $SL(2,\mathbb{Z})$ family of solid-torus fillings, and summing over them fails to produce a positive integer density of states. That is the Maloney–Witten problem, and it is where the course is going.

- *Einstein gravity has not been established.* Sparseness is compatible with a tensionless string in the bulk, as symmetric orbifolds demonstrate. Separating the two needs the gap (15) and strictly sub-Hagedorn light growth (21), neither of which is visible in the leading free energy.

- *Nothing here requires a single* CFT. Everything above is a statement about a leading large-c free energy, which an *ensemble* of theories could reproduce as well as a single one.

Exercise 4. Redo §3.3 with background subtraction instead of the counterterm (57): cut off both geometries at $r = r_c$ and choose the period of the thermal AdS_3 solution so that the two induced boundary metrics agree there. Show that the free-energy difference, and hence the Hawking–Page temperature, are unchanged, but that $M_{AdS_3} = 0$. Explain in CFT language which scheme corresponds to the theory on the cylinder.

Exercise 5. Show that in three bulk dimensions the counterterm series terminates: the only covariant boundary counterterms are $\oint\sqrt{h}$ and $\oint\sqrt{h}\,\mathcal{R}[h]$, and the second is a topological invariant. Then compute $\langle T^i{}_i\rangle$ for a curved boundary metric and fix its coefficient by comparison with the CFT_2 Weyl anomaly $\langle T^i{}_i\rangle \propto c\,\mathcal{R}$; check against [26, 25].

Exercise 6. The extremal limit. Take $r_- \to r_+$ at fixed r_+. Show that $\beta \to \infty$ while $\beta_L = \pi\ell/r_+$ stays finite and $\beta_R \to \infty$, so that only one chirality remains thermally excited, and that $\log Z_{BTZ} \to \frac{c}{24}\cdot 4\pi^2/\beta_L$. Relate this to the Strominger–Vafa configuration $E_L = N_p$, $E_R = 0$ of §1.1 (lecture 3), and note that this is the corner of the BTZ family where (50) is only conjecturally established.

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